

1908 MEASURING RADIATION

In 1907, German physicist Hans Geiger worked with physicist Ernest Rutherford of New Zealand to help him refine his theories about atomic nuclei. In 1908, Geiger made a device to detect radiation (see p.168), later refined and known as the Geiger counter.



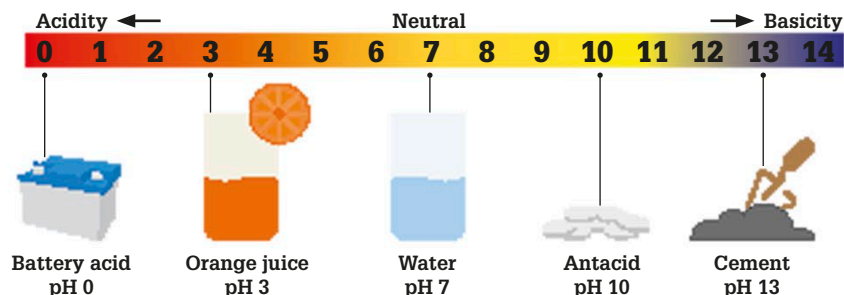
Geiger (left) and Rutherford (right) in the laboratory

Copper tube with gas inside

Early Geiger counter, 1932

Geiger counter

A Geiger counter is a gas-filled tube with a high voltage wire inside. When radiation enters the tube and collides with the gas, it releases electrons that are attracted to the wire. This creates an electric current that produces a clicking sound and interacts with the counter's gauge. The device can detect alpha, beta, and gamma radiation.



1909

Sørensen's pH scale

Devised by Danish chemist Søren Peder Lauritz Sørensen, the pH scale provides an easy way of judging whether a substance is an acid, neutral, or base. The scale runs from 0 (most acidic) to 14 (most basic), with pH 7 indicating the neutral point. Each whole number on the scale is a jump (by 10 times) in the level of acidity or basicity.

1910

Fossil site

In 1909, American paleontologist Charles Walcott discovered the Burgess Shale in the Canadian Rocky Mountains. It is one of the most abundant fossil sites in the world, yielding more than 65,000 fossils since its discovery.

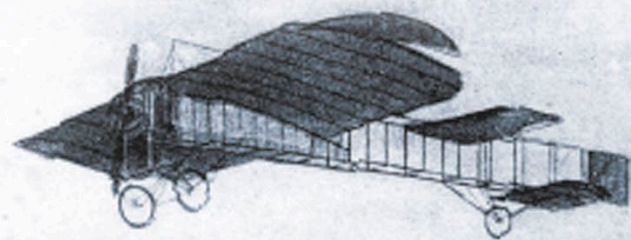


Fossil of a trilobite, a marine animal that lived 526 million years ago

1909

Channel crossing

French aviator Louis Blériot became the first person to fly across the English Channel separating England and France in a heavier-than-air aircraft. His 36-minute, 30-second journey in a 24.91-ft- (7.6-m-) long Blériot XI monoplane helped boost interest in aviation as a practical form of transport.



Blériot crosses the English Channel on his historic 1909 flight



In 1907, American scientist Bertram Boltwood measured the decay of uranium in rocks to calculate their age—an early example of radiometric dating.

